VANDERBILT UNIVERSITY

Department of Biological Sciences

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Search Committee
Department of Anthropology
University of New Mexico

Dear Members of the Search Committee,

I am excited to apply for the position of Assistant Professor of Evolutionary Anthropology at the University of New Mexico. I received my PhD in Biological Anthropology from New York University and am currently in my third year as a postdoctoral researcher in the Department of Biological Sciences at Vanderbilt University. The field, laboratory, and computational components of my research align closely with the Department's interdisciplinary approach to understanding human evolution, integrating ecological, behavioral, and molecular perspectives.

As a biological anthropologist, I study how social environments shape reproduction and fitness across primates. To do so, my research combines sexual selection and life history theory with cutting-edge genomic tools to contextualize reproductive behavior and physiology. During my PhD, I investigated the genetic architecture underlying immune function and mate choice (Petersen et al. 2022, *Ecology and Evolution*), as well as behavioral mating preferences and post-copulatory responses within the female reproductive tract in the olive baboon. This work provided the first evidence of *in vivo* sperm discrimination in a primate species (Petersen et al. 2025 *bioRxiv* in revision at *PLOS Biology*), expanding our understanding of the genetic basis of mate choice and how inter-sexual selection may contribute to infertility in humans. As an NSF postdoctoral fellow, I expanded my genomic expertise to study how social environments shape molecular function. I curated the largest publicly available comparative epigenomics resource to date and examined how early life adversity influences epigenetic aging in rhesus macaques (Sadoughi, Petersen et al., 2025, *bioRxiv* in review at *Science*). I also explored how social environments shape immune responses and life history strategies in rhesus macaques, and cardiometabolic outcomes in humans (Petersen et al., 2025, *bioRxiv* in revision at *Proc R Soc B*). Additionally, I developed wet-lab methods to test causal effects of DNA methylation on gene expression, a potential mechanism through which social experiences can alter physiology (Petersen et al., 2025, *Genome Research*).

Moving forward, I plan to pair my interests in reproductive physiology and sexual selection with functional genomic tools to investigate how social experiences across the life course contribute to reproductive function, evolutionary fitness, and health. I will leverage three complementary systems: free-ranging non-human primate (NHP) populations, human biobanked tissues, and in vitro cell line experiments. During my first five years as a tenure-track scientist, I will advance the following projects:

1. Socioecological drivers of reproductive aging

Primates show striking variation in the timing and extent of reproductive senescence—a phenomenon increasingly recognized as widespread. While ultimate explanations suggest that reproductive senescence may confer inclusive fitness benefits, proximate factors—including early life adversity, energetic trade-offs, and hormonal dynamics—likely shape individual trajectories and influence the evolution of reproductive aging. DNA methylation (DNAm), an epigenetic mechanism regulating gene expression, is a promising pathway linking adversity to variation in physiology due to its environmental responsiveness and its established connections to diverse health outcomes. Through my ongoing collaboration with the Cayo Biobank Research Unit (CBRU) and Dr. Noah Snyder-Mackler (Arizona State University), I am investigating the molecular mechanisms that contribute to female reproductive senescence in humans and NHPs. Using a novel cost-effective method for targeted DNAm sequencing that I helped to develop (Longtin et al. 2025 *PLoS Genetics*), I will generate DNAm profiles from the reproductive tissues of ~100 longitudinally monitored female rhesus macaques and use statistical models to identify adversity associated genomic regions and build epigenetic clocks–a molecular tool that can provide a powerful lens into the pace of tissue aging. In conjunction with long-term demographic and life history information, I will test how the social and environmental exposures—including climate-driven stressors and extreme weather events— shape the pace of aging in female reproductive tissues, evaluating whether adversity accelerates reproductive aging and offering a potential explanation for variation in the timing of reproductive senescence between individuals and species.

2. Linking adversity to complex traits through the epigenome
Adversity profoundly shapes complex traits, including the risk and progression of reproductive diseases such as endometriosis. Epigenetic mechanisms, particularly DNAm, likely mediate these effects, however there is currently limited evidence of how adversity shapes epigenetic variation across the entire genome in reproductive tissues. This leaves a critical gap in our understanding of genome-wide regulatory pathways connecting exposures to complex phenotypes, such as disease. To address this gap, I will generate the first comprehensive map of genome-wide DNAm in endometrial tissue from women with and without endometriosis and perform integrative analyses with disease-associated GWAS variants to pinpoint regulatory elements that drive disease risk. Building on these discoveries, I will leverage data from Study #1 to model whether current and/or early life adversity modifies DNAm at disease-relevant regions in a rhesus macaque model. Finally, I will apply mediation analyses to determine how variation in DNAm translates into cell type–specific changes in gene expression. This project offers a unique opportunity to directly connect the socio-ecological environment to disease-relevant molecular pathways, empirically testing for links between **environment → epigenome → gene expression → complex trait**. My established collaborations with Drs. Digna Velez Edwards and Emily Hodges at Vanderbilt University Medical Center and the Cooperative Human Tissue Network position me to carry out this ambitious work, which will establish a broadly applicable framework for linking adversity to complex traits across species.

3. Expanding functional genomic tools in lesser studied primate species and tissues
A major barrier in primate functional genomics is the lack of species- and tissue-specific resources, which limits our ability to understand how gene regulation evolves across the primate lineage. To address this, I am currently optimizing sequence capture methods for use in massively parallel reporter assays (MPRAs), a wet-lab method which identifies regulatory genomic sequences but has been limited in its application across species due to challenges in cell culture, transfection, and sequence capture methods, which have generally been optimized in humans. This new method would allow for the bioinformatic identification of regulatory sequences across species. In the near term, my future lab will further develop MPRA methods for use in NHPs by optimizing protocols for NHP cell lines, with the long-term goal of expanding NHP cell line models using induced pluripotent stem cells (iPSCs). While iPSCs are often derived from blood or tissue, recent advances in non-invasive iPSC generation could greatly expand their use across difficult to access species and cell types. To advance these methods, I will collaborate with Dr. Genevieve Housman (Max Planck Institute for Evolutionary Anthropology), who has validated protocols for generating, differentiating, and bioinformatically validating NHP iPSCs. Through my training in the New York Consortium in Evolutionary Primatology and my active collaborations with NHP field sites, I have the opportunity to access non-invasively collected samples as starting material for iPSCs from diverse primate species. Together, these resources will enable my lab to investigate how environmental exposures influence gene regulation across species and cell types, providing a framework to understand the evolution of gene regulation and how social and environmental variation shapes molecular evolution across humans and NHPs.

My expertise in evolutionary genomics, functional epigenetics, and primate behavioral ecology, combined with my extensive collaborative network, provides a strong foundation for this proposed integrative work. The breadth of my research program positions me to compete for diverse funding sources supporting anthropology, general biology, and human health research. At the University of New Mexico, I look forward to building collaborations with faculty in the Department of Anthropology–such as Drs. Martin Muller and Melissa Emery Thompson– whose expertise in primate behavioral ecology and reproductive endocrinology would enrich and extend my program. Additionally, I would be ecstatic to continue working with Dr. Ian Wallace, who I have collaborated with during my postdoc to investigate the environmental determinants of health in the Orang Asli of Penninsular Malaysia.

Throughout my career, I have taught a range of courses in biological anthropology, including both laboratory sections and guest lectures within the *Human Evolution* course at New York University and *Introduction to Human Evolution* at Hunter College (CUNY). Guest lectures have included topics such as “Primate Behavior” and “Primate Physiology”, and as a postdoc, I have lectured in Vanderbilt’s graduate-level *Big Data for Biologists* course on “Data Wrangling and Visualization” and “Advanced Data Visualization and Reproducibility” in R. Given the interdisciplinary nature of my work, I am qualified to teach a range of undergraduate-level courses already offered by the department, including *Introduction to Biological Anthropology*, *Human Biology*, and *Human Genetics and Genomics*. I would also be excited to introduce new courses such as *Epigenetics and the Biology of Aging*, *Quantitative Methods in Evolutionary Anthropology*, and *Methods in Evolutionary Genomics*, which could be implemented as lectures or more advanced hands-on computing or wet-lab courses. Lastly, my research program and collaborative field sites will provide excellent training opportunities for students in computational, laboratory, and field-based approaches.

Please find enclosed my CV. Upon request, letters of reference will be provided by: 1) Dr. James Higham, 2) Dr. Amanda Lea, and 3) Dr. Noah Snyder-Mackler. Thank you for considering my application, and please do not hesitate to contact me if additional information is needed.

Sincerely,

RPeter

Rachel Petersen